

DECLINE-CURVE ANALYTICS, MULTI-AGENT AI & GENIE FOR UPSTREAM PRODUCTION

Production Optimizer fuses live well telemetry, decline-curve forecasting, what-if simulation, and a multi-agent supervisor with Databricks Genie to close the gap between forecasted and actual oil production — well by well, hour by hour.

◆ BUSINESS VALUE

- ▶ **Close the forecast gap** — quantify deviation from decline-curve baselines and act before barrels are lost
- ▶ **Optimize per-well intervention** — choke, ESP frequency, and gas-lift sliders with simulated NPV impact
- ▶ **Natural-language exploration** — Ask Genie answers ad-hoc questions over Unity Catalog production tables
- ▶ **Multi-agent orchestration** — Supervisor agent routes between monitoring, optimization, and recommendation specialists
- ▶ **One pane of glass** — geospatial field map, digital twin, deep-dive analytics, and AI advisors in a single app

◆ CHALLENGES ADDRESSED

- ▶ **Forecast vs. actual blindness** — decline curves live in spreadsheets, decoupled from live SCADA
- ▶ **Slow what-if cycles** — engineers can't simulate choke / lift changes without re-running offline models
- ▶ **SQL gatekeeping** — operations leaders wait days for production-data answers from analysts
- ▶ **Recommendation overload** — multiple ML models surface alerts with no prioritization or business context
- ▶ **Disconnected analytics** — map, twin, and KPI dashboards live in separate, non-interactive tools

◆ WHO IT'S FOR



Production Engineer

Per-well decline-curve tracking, choke & lift optimization, recommended actions



Operations Manager

Field-wide KPI ticker, alert prioritization, NPV impact of interventions



Reservoir Analyst

Ask Genie for ad-hoc NL queries across production, pressure, and economic tables



Asset Director

Aggregate field economics, barrel-shortfall accountability, capital allocation

6

Integrated Tabs

5

AI Sub-Agents

1

Genie Space

3s

Live Refresh

◆ QUICK START GUIDE

1

Clone Repository

Clone the Production Optimizer repo to your workspace

2

Bootstrap Schema

Run setup SQL to seed wells, telemetry, decline-curves into Unity Catalog

3

Deploy the App

Run `databricks apps deploy` with your warehouse + Genie space IDs

4

Optimize & Ask

Open the app, pick a well, simulate a choke change, then Ask Genie what changed

◆ GETTING THE MOST OUT OF PRODUCTION OPTIMIZER



Field Overview

Geospatial map of all wells, producers, and injectors with live status + KPI ticker



Decline Deep Dive

Per-well forecasted vs. actual curves with health bars and ranked underperformer list



Optimize Sub-Tab

Slider-driven what-if for choke size, ESP frequency, and gas-lift with instant NPV impact



Ask Genie

Natural-language SQL over production tables, governed by Unity Catalog and grounded by your Genie space

◆ TARGET INDUSTRY

Upstream Oil & Gas — Production engineering and operations on conventional and unconventional onshore assets. Applicable to any operator tracking decline-curve performance, artificial-lift optimization, and per-well economics.

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